

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

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Submitted to:

Ms. Saadia Shabbir

Submitted by:

Zulqarnain Hassan

Roll No: SP21379

Subject: Web engineering

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Lab Report: Pizza Ordering Web Application

1. Introduction

The objective of this lab was to create a functional pizza ordering web application using HTML, CSS, JavaScript, and PHP. This application simulates a real-world pizza-ordering scenario, allowing users to select pizzas, add extra toppings, calculate prices dynamically, apply discounts, and generate an order receipt.

2. Implementation Details

2.1 HTML (pizza-order.html)

The HTML file is the backbone of the application, containing the structure and layout of the web page. Key elements included:

- **Header Section:** Displaying the application title and navigation (if applicable).
- **Pizza Selection:** A section with buttons or an auto-slide feature to showcase available pizzas. Hover effects were added to display brief descriptions.
- **Order Form:** Inputs for selecting pizzas, extra toppings, delivery options, and entering a discount code.
- **Submit Button:** A button to place the order and trigger the server-side processing.

2.2 CSS (pizza-order.css)

The CSS file was used to style the application, focusing on:

- **Layout Design:** Arranging elements in a user-friendly and visually appealing manner.
- **Hover Effects:** Implementing tooltips over pizza names to display descriptions.
- **Dynamic Styling:** Adding a glowing effect to the price calculation box whenever the total price updates.
- **Responsive Design:** Ensuring the web page is accessible on various devices.

2.3 JavaScript (pizza-order.js)

JavaScript added interactivity and dynamic features:

- **Pizza Selection and Price Calculation:** Event listeners detect selections and update the total price in real time.
- **Topping Selection:** Prices increase or decrease based on user choices.
- **Delivery Options:** Adding \$5 for home delivery or no charge for pick-up.
- **Discount Code:** Validating a discount code (e.g., DISCOUNT10) to apply a 10% discount or display an error message for invalid codes.
- **Error Handling:** Displaying warnings if no pizzas are selected when attempting to place an order.

2.4 PHP (pizza-order-processing.php)

The PHP file handled server-side processing:

- **Form Validation:** Checking that all required fields were completed and inputs were valid.
 - **Order Processing:** Generating an HTML receipt summarizing the order, including:
 - Selected pizzas and toppings.
 - Delivery method.
 - Applied discounts.
 - Final price breakdown.
 - **Error Handling:** Returning appropriate messages for invalid inputs or failed processing.
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3. Challenges and Solutions

- **Real-Time Updates:** Ensuring price calculations reflected all user selections dynamically required precise event handling.
 - **Solution:** Implemented JavaScript event listeners and DOM manipulation techniques effectively.
 - **Discount Validation:** Preventing invalid discount codes from affecting the total price.
 - **Solution:** Added client-side and server-side validation.
 - **PHP Integration:** Handling form data securely and generating dynamic responses.
 - **Solution:** Used PHP functions like `filter_input` and `htmlspecialchars` to sanitize inputs.
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4. Results

The following features were successfully implemented:

- A visually appealing interface for pizza selection.
- Real-time price updates with glowing effects.
- Functional discount code validation.
- Clear error messages and warnings.
- A dynamic, server-generated order receipt.

Screenshots:

Welcome to Joe's Pizza!

You order, we deliver.

Place your order now!

Please Fill The Form Below.

Pizza Order

Choose Your Pizza

Vegetarian	☒ Vegetarian (\$12, 6-inch)	☐ Vegetarian (\$20, 12-inch)
Margherita	☐ Margherita (\$5, 6-inch)	☒ Margherita (\$8, 12-inch)
Bufala	☒ Bufala (\$16, 6-inch)	☐ Bufala (\$25, 12-inch)

Extra topping?

☐ Mushroom.....\$6

☒ Mozzarella Cheese.....\$5

☐ Meat.....+\$7

Delivery Options:

☐ Pick-Up

☒ At-Home Delivery (+\$5)

Total Price:

\$46.00

Contact Information:

Name:

zulqarnain Hassan

Phone:

03436522750

Address:

humi internal hotel-4

Place Order

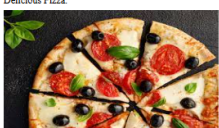
Place Order



Pizza made with love.



Delicious Pizza.



Delicious Pizza.

Order Summary

Dear zulqarnain Hassan, we have received the following order from you:

Vegetarian 6-inch

Margherita 12-inch

Bufala 6-inch

With the following extra toppings: Mozzarella Cheese

Total price is \$46.00

Thank you for ordering from us!

We will be on your way soon.

5. Conclusion

This lab provided practical experience in developing a fully functional web application, integrating front-end and back-end technologies. It enhanced skills in HTML, CSS, JavaScript, and PHP, offering insights into real-world web development scenarios.